

**A PROPOSED OFFERING OF
A VILLAGE RESIDENT INFORMATION & PROFILING SYSTEM
FOR DISIPLINA VILLAGE, UGONG**

A Project Proposal Presented to the
Faculty of Datamex College of Saint Adeline, Inc.

In Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Information Technology

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REQUIREMENTS SPECIFICATION

INTRODUCTION

This document is created to clearly explain what the Village Resident Information & Profiling System will do and what is expected from it. It serves as a formal reference for project stakeholders including Village office officials, system developers, and end-users to ensure a shared understanding of the system's features, objectives, and constraints. This document will guide the development process, minimize misunderstandings, and provide a basis for validation and testing of the final product.

The Village Resident Information & Profiling System is designed to make managing resident and household records faster, easier, and more reliable. Instead of relying on manual methods like logbooks, index cards, or scattered Excel files, this system provides one secure platform where officials can add and edit resident profiles, group family members into households, monitor population growth, and . It also has built-in security features, such as login and user , to ensure that sensitive information is protected. Overall, the system helps office staff work more efficiently while reducing errors and duplication.

This document covers both the features the system should have and the standards it needs to follow. It will describe the functions like creating and updating resident profiles, managing households, and generating statistics, as well as non functional aspects like security, ease of use, and performance. The system is mainly for village office officials and staff of Disiplina Village, Valenzuela City, but residents also benefit indirectly because their information will be handled more securely and services will be delivered more quickly. This requirements specification will act as the foundation for designing, building, and testing the system.

REQUIREMENT GATHERING

The requirement gathering table outlines the key features and functions of the Village Resident Information & Profiling System. It defines how the system should handle admin login, resident and household management, data security, and dashboard statistics. Each requirement is prioritized, linked to dependencies, and includes acceptance criteria to ensure proper implementation and testing.

Requirement ID	Requirement Description	Priority	Dependencies	Acceptance Criteria
FR-01	The system shall allow the admin to log in using a username and password.	High	Database connection	Admin enters valid credentials → Access to Dashboard is granted.
FR-02	The system shall restrict access to authorized admin accounts only.	High	FR-01	Invalid credentials → Access is denied with an error message.
FR-03	The system shall display a Dashboard with statistics (residents,	Medium	Resident and household data	Dashboard shows updated counts and charts after

	households, gender, age groups).			data entry.
FR-04	The system shall allow the admin to add new residents with fields (Full Name, Age, Gender, DOB, Address, Household ID, etc.).	High	Database connection	Resident record is saved and appears in View Residents table.
FR-05	The system shall allow the admin to edit or delete existing resident records.	High	FR-04	Changes reflect immediately in the residents table.
FR-06	The system shall provide a search and filter function in the View Residents module.	Medium	FR-04	Admin can quickly locate residents by name, household, or other criteria.
FR-07	The system shall allow the admin to create and	High	FR-04	Household data saved and

	manage households (Household ID, Head, Members, Address).			residents can be assigned correctly to a household.
FR-08	The system shall allow assigning residents to specific households.	High	FR-07	Resident list updates to show correct household assignment.
FR-09	The system shall allow the admin to update account credentials in Settings.	High	FR-01	Admin credentials updated and must be used on the next login.
FR-10	The system shall allow the admin to log out securely and return to the Login screen.	High	FR-01	User session ends, returning to Login form.

NON FUNCTIONAL REQUIREMENT

The non-functional requirements define the system's quality standards beyond its core features. They ensure the platform is secure, reliable, and easy to use while supporting scalability, performance, and data integrity. These requirements also cover maintainability and backup measures to guarantee long-term stability and smooth operation.

Category	Requirement Description
Performance	The system should load the Dashboard and Resident records within 3–5 seconds under normal usage. Database queries should return results efficiently even with 1,000+ records.
Usability	The system shall provide a clean and intuitive user interface designed in Visual Basic .NET, ensuring that admin users can easily navigate and manage residents with minimal training.
Reliability	The system should ensure 99% uptime during operation (excluding maintenance) and handle unexpected input errors gracefully without crashing.
Security	The system shall enforce admin-only access with secure password storage (e.g., hashing). Unauthorized users must not gain access to resident or household data.
Scalability	The system should support future expansion, including the ability to manage up to 5,000 resident records without performance degradation.
Maintainability	The system code should be modular and well-documented, allowing easy updates, bug fixes, and enhancements in Visual Studio.

Data Integrity	All resident and household data must remain consistent and protected against accidental deletion or corruption.
Backup & Recovery	The system shall allow regular database backups through SQL Server and ensure recovery in case of data loss or corruption.

USE CASES

The use cases describe the main interactions between the **admin**, **staff**, and the **system**, such as logging in, adding resident profiles, managing households, and generating reports. Each use case outlines the steps, conditions, and alternate flows to ensure smooth operation and error handling. Together, they define how the system supports barangay officials and staff in managing resident and household information effectively.

Use Case ID	UC-001
Use Case Name	Admin Login
Description	The admin provides login credentials to access the system.
Actors	Admin
Preconditions	Admin account must exist in the system.
Postconditions	Admin is authenticated and redirected to the dashboard.
Main Flow	1. Admin enters username and password. 2. System validates credentials 3. If valid, system logs the admin in and opens the dashboard.

Table 1: The admin securely logs in with valid credentials to access the system dashboard.

Use Case ID	UC-002
Use Case Name	Dashboard
Description	The system displays an overview of records such as total residents, total households, and archived records.
Actors	Admin
Preconditions	Admin must be logged in.
Postconditions	Admin views the dashboard and may navigate to other modules.
Main Flow	1. Admin logs in successfully. 2. . System shows key statistics and summaries. 3. . Admin navigates as needed.

Table 2: The admin registers a new resident by entering complete personal and household details into the database.

Use Case ID	UC-003
Use Case Name	Manage Household
Description	The admin groups residents into households, assigning or updating a Household ID.
Actors	Admin
Preconditions	At least one resident must exist in the system.
Postconditions	Residents are correctly grouped under the assigned Household ID.
Main Flow	1. Admin selects Manage Household. 2. System displays list of residents. 3. . Admin assigns/updates Household IDs. 4. . System saves changes. 5. . Confirmation displayed.

Table 3: The admin groups or updates residents under a specific household for organized profiling.

Use Case ID	UC-004
Use Case Name	Add Resident Profile
Description	The admin records a new resident's personal and household information into the system.
Actors	Admin
Preconditions	Admin must be logged in.
Postconditions	New resident profile is saved in the system.
Main Flow	1. Admin selects Add Resident Profile. 2. . System shows registration form 3. Admin enters resident details 4. System validates inputs. 5. Data saved. 6. Confirmation displayed.

Table 4: The system generates and displays reports such as population count, household data, and voter status.

Use Case ID	UC-005
Use Case Name	Archive
Description	The admin archives resident or household records for historical tracking instead of permanent deletion.
Actors	Admin
Preconditions	Admin must be logged in.
Postconditions	Selected records are archived but retrievable later.
Main Flow	1. Admin selects a record. 2. . System moves the record to the archive 3. System confirms archiving.

Table 5: The admin securely logs out, ending the session and returning to the login page.

Use Case ID	UC-006
Use Case Name	Admin Logout
Description	The admin clicks the “Logout” button to securely exit the system. The session ends, and the system returns to login.
Actors	Admin
Preconditions	Admin must be logged in.
Postconditions	System ends the session, clears login credentials, and redirects to the login screen.
Main Flow	1. Admin clicks Logout 2. System ends session. 3. Admin is redirected to login page.

Table 6: Allows the admin to securely exit the system, ending the session and returning to the login screen.

DATA REQUIREMENTS

DATA ENTITIES

- **Admin** – Stores administrator login credentials and information.
- **Login Module** – Handles admin authentication.
- **Dashboard** – Displays system population statistics.
- **Resident Management** – Allows the admin to View, Edit, and delete resident records.
- **Household Management** - Enables the admin to group residents into households with a unique ID and manage details like head and members for organized profiling.
- **Archive** - Keeps deleted resident and household records in secure storage, allowing the admin to restore them when needed.
- **Logout** – Securely ends the admin session.

Staff Account Component

The Staff Account allows authorized personnel to assist the admin in managing resident and household information. Their access is limited but essential to maintain system efficiency and data accuracy.

• Dashboard

- Displays an overview of key system data such as total residents, households, and recent activities.
- Provides quick access to commonly used functions like viewing resident lists or household records.
- Shows staff notifications, assigned tasks, and updates from the admin.
- Helps staff monitor ongoing activities, such as scheduled interviews or community programs.

• Resident Management

- Allows staff to **view and update** resident information when authorized by the admin.
- Enables staff to **add new resident records** or **edit details** like address, contact number, or occupation.
- Includes a **search and filter** function for quick access to resident profiles.
- Helps ensure resident data remains **accurate and updated** for barangay reports and services.

• Household Management

- Lets staff **view household groups** and see which residents belong to each family.
- Enables updating of **household information**, such as the head of the family or household address.
- Allows staff to assist in tracking **household-based programs** like aid distribution or community visits.

- Promotes better organization and understanding of **family structures** within the barangay.
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Statistics / Reports (Derived Data) – Generated from resident and household data for dashboard display.

ATTRIBUTES

Admin

- Username
- Password

RESIDENT

- ResidentID (Primary Key)
- FullName

- Gender
- Age
- Date of birth
- Address
- Contact
- Civil Status
- Voter Register
- HouseholdID (Foreign Key)

HOUSEHOLD ROLE

- HouseholdID (Primary Key)
- Head Of the family
- Spouse
- Children
- Parents
- Relative
- Others

STATISTICS / REPORTS (DERIVED, NOT STORED DIRECTLY BUT COMPUTED)

- TotalResidents
- TotalHouseholds
- TotalVoters

RELATIONSHIPS

One-to-Many (Household → Residents):

- Each household can have multiple residents.
- Example: HouseholdID = H1 may have 5 residents linked to it.

One-to-One (Admin → System):

- Only the Admin manages the system.

Derived Relationships (Statistics):

- Reports are generated by counting or grouping data from Resident ,Household and Voter Status tables

ASSUMPTIONS AND CONSTRAINTS

- The system will be used only by authorized administrators; there will be no resident self-service portal.
- Administrators are expected to have basic computer knowledge to navigate the system effectively.
- The database (SQL Server) will run on a stable local server or school-provided machine without frequent downtime.
- Resident and household information provided by the admin is assumed to be accurate and updated at the time of entry.
- The system will be primarily designed for use within the local barangay office or institution, not over the internet.

CONSTRAINTS

- The system is developed using Microsoft Visual Basic 2010 (VB.NET) and SQL Server (SSMS), so it is dependent on these platforms.
- Since it is a desktop-based system, it will not be accessible on mobile devices or browsers.
- System access is limited to admin accounts only, meaning no public user access is available.
- The performance of the system is constrained by the hardware capacity of the computer where it is installed (e.g., memory, storage).
- Future scalability is limited; if the barangay population grows significantly, database optimization may be required.

GLOSSARY

Administrator (Admin)

The administrator is the main user of the system with full control over everything. They handle resident and household records, making sure all information is accurate and up to date. Only the admin can log in, approve, or update data. Without the admin, the system can't function since all key actions depend on their authority.

Dashboard

The dashboard acts like the system's control panel after login. It gives the admin quick access to modules such as Resident Management, Household Management, Archive, and Reports. It also shows important statistics like the total number of residents and households. With this, the admin can easily keep track of the village office overall status.

Resident

A resident is anyone living in the village office whose details are stored in the profiling system. Each resident has unique information such as their name, age, and address. This data helps track population numbers and organize households. With accurate resident records, the village office can better understand and respond to community needs.

Household

A household is a family or group of people living together in one home. Each household has a unique Household ID that connects its members in the system. This makes it easier to see family relationships, track household size, and organize services for the community. It also helps identify the head of the household and how resources are shared.

Archive

The archive is where old, inactive, or deleted records are safely stored. Instead of being permanently erased, records are kept here for reference or future use. This makes sure resident and household information can still be traced or restored when needed. Archiving also helps keep the system neat by separating active records from old ones.

Logout

The logout function lets the admin safely exit the system after using it. Once logged out, the session ends and access to sensitive data is blocked. This keeps the system secure and prevents unauthorized access. Logging out is an important step to protect information from being misused.

SSMS (SQL Server Management Studio)

SSMS is the database management tool used for handling all stored information in the system. It ensures that resident and household records are saved securely and can be retrieved when needed. Using SSMS, the admin can also perform

database queries, backups, and maintenance. It is the backbone of the system's data handling and ensures data integrity.

Microsoft Visual Basic 2010 (VB.NET)

Microsoft Visual Basic 2010 is the development environment used to build the profiling system. It provides the tools needed for coding, debugging, and designing the user interface. This makes it possible to connect the system to SSMS for smooth data transactions. The combination ensures that the system is reliable and user-friendly for the admin.

Login

Login is the process where the administrator enters their username and password to access the system. It ensures that only authorized personnel can use the application. Without a successful login, no data can be viewed, added, or modified. This provides security and protects sensitive information from unauthorized access.

Profiling

Profiling is the detailed process of collecting, organizing, and storing resident information. It includes personal data, household relationships, and demographic details. This data is later used for reports, statistics, and decision-making within the barangay. Profiling ensures that every resident is properly recorded for community planning and management.

REVISION HISTORY

The revision history keeps track of changes made to this requirements document. For now, only the initial draft has been recorded since no updates or modifications have been made yet. Future changes will be logged here to ensure proper version control and traceability.

Table: Revision History

Version	Date	Author	Description of Change
1	August 13, 2025	Gabriel Thomas Torneros	Revised
2.	August 20, 2025	Gabriel Thomas Torneros	Revised
3.	August 28,2025	Gabriel Thomas Torneros	Format

APPENDIX

The appendix provides supporting materials that complement the main content of the study. It includes mockups of the system interface, diagrams such as ERD and DFD, sample data used for testing, user stories based on Agile methodology, and details of the backup and recovery plan. These documents serve as references to better understand the system's design, functionality, and implementation.

MOCKUPS

- Login Screen
- Two text fields (Username, Password) with “Login” button.
- Error label for invalid credentials.

Dashboard:

- Displays statistics such as total residents, total households, Total Of Register Voters Status.
- Sidebar navigation buttons Dashboard, Resident , Household,Archive Logout.

Add Resident Form:

- Input fields for Full Name, Gender, Age, Date of Birth, Address, Contact, Civil Status, Voter Register, Household ID Household Role, Household ID, Head Of The Family, Spouse, Children, Parents, Relative, Others.
- Buttons: Add New Resident, Add New Household, Cancel, View, Edit, Delete

View Resident Information:

- Table listing: Resident ID, HouseholdID, FullName, Gender, Age, Date Of Birth, Civil Status. Voter Register, HouseHold Role, ContactNumber, Address.
- Buttons: Add NewHousehold, Add New Resident, View,Edit,Delete.

DIAGRAMS

Entity-relationship diagram (erd):

Resident

- ResidentID
- FullName
- Gender
- Age
- Date of birth
- Address
- Contact
- Civil Status
- Voter Register
- HouseholdID (Foreign Key)
- Head Of the family
- Spouse
- Children
- Parents
- Relative
- Others
- **ViewResident Information** (Resident ID, HouseholdID, FullName, Gender, Age, Date Of Birth, Civil Status, Voter Register, HouseHold Role, ContactNumber, Address)

- **Admin**

One Household → Many Residents; Admin → manages Residents & Resident Information.

- **Data Flow Diagram (Level 0):**

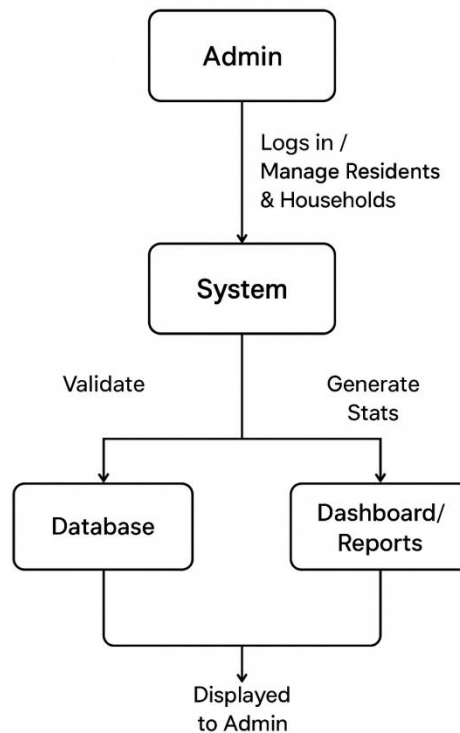
Admin logs in → System → Validates via Database.

- **Admin** manages residents (Add/View/Edit/) → Updates Database.
- **Admin** manages households (Delete,Cancel) → Updates Database.
- **System** provides dashboard statistics → Displays to Admin.

Appendix A.

The Level 0 Data Flow Diagram (DFD) provides a high-level view of how the Admin interacts with the Village Resident Information & Profiling System. It highlights the main processes, data flows, and interactions between the Admin, the System, and the Database.

Appendix 1 Use Case Diagram



The DFD shows that the Admin can log in, manage residents and households, view the dashboard, and log out. The System validates and updates the Database, generates statistics and reports, and displays them to the Admin, ensuring secure, efficient, and controlled access to all resident data.